

Xiaoqing PAN

xiaoqing.pan@uni-jena.de | [LinkedIn](#) | [ORCID](#) | [Website](#) | +49 15563309561 | Jena, Germany

MACHEREY-NAGEL Vertrieb GmbH & Co. KG - Valencienner Str. 11 - 52355 Düren
University Hospital Muenster

Re: Research Associate // Job ID: 432WM-B // reference number: 11536

Dear Hiring Team,

I am writing to express my interest in the research associate position for Research & Development in Bioanalytics. As a doctoral candidate in microbiology at the Hans-Knöll-Institute in Jena, Germany (thesis submitted in Jan 2026), my research mainly focused on double-stranded RNA (dsRNA) metabolism in host-fungal interactions.

My journey in microbiology started by characterizing the RNA interference (RNAi) pathway in the human pathogen *Aspergillus fumigatus*. By investigating sRNA landscape in *A. fumigatus*, we have found out that the fungus produced canonical sRNA under stress growth environments. Besides that, I have also worked on host-fungal interactions by investigating how human A-to-I RNA editing and RNA splicing patterns change upon fungal stimulations where I mainly contributed to bioinformatic analysis.

Based on the job description, the prerequisites align closely with my background in molecular biology. I have in-depth knowledge of DNA/RNA/protein. During my PhD, I developed methods for fungal studies, including DNA/RNA isolations.

So far, my work has primarily focused on computational approaches for RNA-seq analysis, and some hands-on experimental work, including cell culture growth assays, Northern Blot and PCR techniques. This position will provide me with the opportunity to become someone who can generate hypotheses and experimentally test them. In the future, I hope to contribute to the medical industry, and working as part of this fascinating project would present a great stepping stone to gain broader expertise in the research of diseases beyond fungal pathogens.

My PhD supervisor, Dr. Matthew Blango (matthew.blango@leibniz-hki.de) and Dr. Olaf Kniemeyer (olaf.kniemeyer@leibniz-hki.de) have agreed to serve as referees. Salary expectation is 60000 euros annually. And the earliest date can be 01.02.2026. Thank you for your time and consideration.

Best,

Xiaoqing

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SUMMARY

Doctoral candidate in **Microbiology** specializing in RNA biology in the context of host-pathogen interactions. Proficient in R, Linux and NGS data processing within HPC environments (Slurm). Experienced in RNA-seq analysis, prediction of A-to-I RNA editing and alternative splicing and microbial assays. Passionate about integrating bioinformatics to address questions in biotechnology and immunology. Published five papers and two in preparation.

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany 2026
Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs

MSc | Bioengineering, Chinese Academy of Sciences, China 2021
Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively

PROFESSIONAL EXPERIENCES

Human PBMC transcriptome changes upon fungal stimulations

- Designing a high-throughput **Snakemake workflow** to quantify differential expression and alternative splicing events, characterizing the dynamic response of human PBMCs to fungal pathogens.
- Leading a six people-involved project. Collaborate with Fritz-Lipmann-Institut, Germany
- DOI: [10.1021/acsinfecdis.4c00598](https://doi.org/10.1021/acsinfecdis.4c00598)

RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Develop both sRNA- and mRNA-seq pipeline for differential expression gene analysis.
- Experimental validation support for determining the role of RNAi machinery and sRNA using Northern Blot, PCR, qPCR assays.
- Leading and involving a twenty people project. Collaborate with The University of Manchester, the University of Jena
- DOI: [10.1261/rna.079350.122](https://doi.org/10.1261/rna.079350.122) and DOI: [10.1101/2024.05.24.595671](https://doi.org/10.1101/2024.05.24.595671)

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analysis of human extracellular small RNA transcriptome.
- Involved in a ten people project. Collaborate with the University of Jena.

Non-coding RNA analysis of rice-*Xoo* interaction

- Experimental validation support for determining non-coding RNAs (sRNAs, lncRNAs, circRNAs) in rice responding to fungal infections using PCR, qPCR, LC-MS, assays.
- Leading a five people-involved project. Collaborate with Sun Yat-Sen University, China
- DOI: [10.1186/s42483-023-00188-8](https://doi.org/10.1186/s42483-023-00188-8) and DOI: [10.1016/j.jplph.2022.153905](https://doi.org/10.1016/j.jplph.2022.153905)

Teaching

- Microbiology and Molecular Biology - MMB005, 2022 - 2025
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TECHNICAL SKILLS

- **Bioinformatics:** HPC, Snakemake, Genomics, RNA-seq, SHAPE-seq, Alternative splicing, Data visualization
 - **Molecular Biology:** DNA/RNA/protein isolation, Genetic Engineering, Northern Blot, qPCR, PCR, LC-MS, Gibson Assembly Cloning, Cultivation of microorganisms
 - **Programming & Tools:** R, python, Bash, Git, Docker, Linux
 - **Language:** English (fluent), German (B1), Chinese (native)
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PUBLICATIONS

Pan X*, Kelani AA*, et al. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microb.

Pan X*, et al. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis

10(12), 4017–4029.

Xia K*, Pan X*, *et al.* (2023). X00-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of OsARAB in rice. *Phytopathol Res* 5, 22.

Xia K*, Pan X*, *et al.* (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting OsOFP3, OsNPF2.4, and OsAGO1a, respectively. *J. Plant Physiol* 208, 153905.

Kelani AA, ..., Pan X, ..., Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. *RNA* 29(7), 1033–1050.

CONFERENCES

2025 - Oral presentation: German speaking Mycological Society (DMykG), Germany

2025 - Poster: Gordon Research Seminar - RNA Editing, Italy

2024 - Oral presentation: Jena RNA Club Symposium, Germany

2024 - Poster: RNA Society Meeting, UK // Poster

2023 - Poster: EMBL - The Non-coding Genome Symposium, Germany

AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
- Jena School for Microbial Communication (JSMC) travel grant 2023